

Automated EDA using Pandas Profiling

100 Days of Machine Learning | Day 22 | CampusX

1. What is Pandas Profiling?

In previous steps of Exploratory Data Analysis (EDA), we manually asked basic questions, performed Univariate Analysis, and conducted Bivariate/Multivariate Analysis. **Pandas Profiling** (now widely known as `ydata-profiling`) is a powerful Python library that automates this heavy lifting.

With just a few lines of code, it generates a comprehensive, interactive HTML report covering almost everything you need to know about your dataset.

2. Installation and Usage

Installation

Open your terminal or Jupyter Notebook cell and run:

```
!pip install pandas-profiling
```

(Note: In modern environments, the library is updated and often installed via `!pip install ydata-profiling`)

Implementation

```
import pandas as pd
from pandas_profiling import ProfileReport

# 1. Load the dataset
df = pd.read_csv('titanic.csv')

# 2. Create the Profile Report object
prof = ProfileReport(df)

# 3. Export the report to an HTML file
prof.to_file(output_file='output.html')
```

Once you run this code, an HTML file (e.g., `output.html`) is generated in your directory. You can open it in any web browser to view the interactive dashboard.

3. Sections of the Profiling Report

The generated HTML report is divided into several incredibly useful sections:

A. Overview

This section answers the "Basic Data Questions":

- **Dataset Statistics:** Total number of variables (columns), number of observations (rows), duplicate rows, and total missing cells.
- **Variable Types:** How many columns are numerical vs. categorical.
- **Warnings (Crucial):** It automatically alerts you to potential issues like *High Cardinality* (too many unique categorical values), high percentages of *Missing Values* (e.g., 77% missing in the Cabin column), or columns that have predominantly zero values.

B. Variables (Univariate Analysis)

It provides an in-depth univariate analysis for every single column:

- **Categorical Data:** Shows countplots, pie charts, distinct counts, missing values, and length details.
- **Numerical Data:** Displays a histogram, the 5-number summary (Min, Q1, Median, Q3, Max), Interquartile Range (IQR), standard deviation, skewness, variance, and explicitly flags outlier values.

C. Interactions (Bivariate Analysis)

This section allows you to interactively plot any numerical column against another numerical column. It automatically generates scatter plots, helping you spot linear or non-linear relationships instantly without writing plotting code.

D. Correlations

Provides visual correlation matrices (Heatmaps) to show how strongly features are related to one another. It includes different correlation coefficients like:

- **Pearson:** Measures linear correlation.
- **Spearman:** Measures monotonic relationships.

E. Missing Values

Visualizes missing data through Bar charts and Matrix plots. The matrix plot helps you identify if the missingness in one column is correlated with the missingness in another column.

F. Sample

Displays the `head()` (first few rows) and `tail()` (last few rows) of your dataset so you can visually inspect the raw data.

4. Best Practice

Pandas Profiling is a tool for **Self-Exploration**. Whenever you get a new dataset, run this tool first. Spend time analyzing the generated HTML report and noting down observations. It will save you hours of manual plotting and guide your feature engineering steps.